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324553(24)

BE (5th Semester)
Examination, Nov.-Dec., 2015

(New Scheme)

Applied Numerical Analysis

Time Allowed : 3 hours *Maximum Marks* : 80
Minimum Pass Marks : 28

Note : (*i*) Part *(a)* of each question is compulsory and attempt any two parts of remaining *(b)*, *(c)* and *(d)*.

(*ii*) The figures in the right-hand margin indicate marks.

1. *(a)* Define transcendental equation. 2

(b) Find the roots of the equation $xe^x = \cos x$ using the secant method correct to four decimal places. 7

(c) Solve by Newton-Raphson method $x^3 - 3x + 1 = 0$, correct to three decimal places. 7

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(d) If $u = \dfrac{4x^2y^3}{z^4}$ and errors in x, y, z be 0·001, compute the relative maximum error in u when $x = y = z = 1$. 7

2. *(a)* What is the condition of failure of Gauss elimination method? 2

(b) Apply Gauss-Jordan method to solve the equations : 7

$$x + y + z = 9$$
$$2x - 3y + 4z = 13$$
$$3x + 4y + 5z = 40$$

(c) Solve

$$5x - y + z = 10$$
$$2x + 4y = 12$$
$$x + y + 5z = -1$$

correct to three decimal places by Jacobi's iteration method. Start with the solution (2, 3, 0). 7

(d) Find the eigenvalues and eigenvectors of the matrix : 7

$$A = \begin{bmatrix} 8 & -6 & 2 \\ -6 & 7 & -4 \\ 2 & -4 & 3 \end{bmatrix}$$

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3. *(a)* Define interpolation. 2

(b) Find the least squares approximation of second degree for the discrete data : 7

x :	–2	–1	0	1	2
$f(x)$:	15	1	1	3	19

(c) By means of Lagrange's formula, prove that

$$y_3 = 0{\cdot}05(y_0 + y_6) - 0{\cdot}3(y_1 + y_5) + 0{\cdot}75(y_2 + y_4)$$ 7

(d) The following values of the function $f(x)$ for the values of x are given

$$f(1) = 4,\ f(2) = 5,\ f(7) = 5,\ f(8) = 4$$

Find the value of $f(6)$ by means of Newton's divided difference formula. 7

4. *(a)* Evaluate $\Delta \tan^{-1} x$. 2

(b) A curve is drawn to pass through the points given by the following table :

x :	1	1·5	2	2·5	3	3·5	4
y :	2	2·4	2·7	2·8	3	2·6	2·1

Estimate the area bounded by the curve, the x-axis and the line $x = 1$ to $x = 4$ by Simpson's one-third rule. 7

(c) Calculate the first and second derivatives of the functions tabulated below at the point $x = 1{\cdot}1$: 7

x :	1·0	1·2	1·4	1·6	1·8	2·0
$y = f(x)$:	0	0·128	0·544	1·296	2·432	4·000

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(d) A slider in a machine moves along a fixed straight rod. Its distance x cm along the rod is given below for various values of the time t seconds. Find the velocity and acceleration of the slider when t = 0·3 second. Use Stirling's formula :

t :	0	0·1	0·2	0·3	0·4	0·5	0·6
x :	30·13	31·62	32·87	33·64	33·95	33·81	33·24

5. *(a)* Write Milne's predictor-corrector formula. 2

(b) Find, by R-K method, an approximate value of y for $x = 0{\cdot}8$, given that $y = 0{\cdot}41$ when $x = 0{\cdot}4$ and

$$\frac{dy}{dx} = \sqrt{x+y}$$

(c) Use modified Euler's method to compute y for $x = 0{\cdot}05$, given that $\frac{dy}{dx} = x + y$ with initial condition $x_0 = 0$, $y_0 = 1$ correct up to three decimal places. 7

(d) Using Taylor's series, find the solution of the differential equation $xy' = x - y : y(2) = 2$ at $x = 2{\cdot}1$, correct to five decimal places.

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